What the transformer evidence adds

Optional debrief · only after Notebook 2 CORE 5 · 3–5 minutes

ILIAD Intensive · Computational Mechanics

Mess3: compelling geometry, incomplete discrimination

Shai et al. report that an affine map from 64-dimensional final residual activations:

Recovers geometry

Decoded points reproduce the three-state belief structure.

Tracks training

Regression error decreases as the model learns.

Survives controls

Held-out MSE remains similar under random 20/80 input–activation-pair splits; shuffled correspondence collapses.

But on Mess3, next-token probabilities already track much of the belief geometry. The example alone cannot isolate richer predictive state.

Reported results: Shai et al., NeurIPS 2024, §§3.1 and A.7. The random split and shuffle procedures were independently repeated 1,000 times.

RRXOR supplies the matched-prediction test

RRXOR has 36 belief states, including histories with matching optimal next-token distributions.

Layer pattern

Final and individual-layer probes recover the geometry poorly; a probe over concatenated layers recovers it much better.

Diagnostic comparison

Recovered distances can be compared separately with belief distances and with optimal next-token-distribution distances.

$$R^2 = 0.95$$

relation to ground-truth belief distances

$$R^2 = 0.31$$

relation to optimal next-token-distribution distances

Strong observational evidence, one rung still open

Supported

- affine decodability;
- training-linked geometric structure;
- distinctions beyond current optimal next-token probabilities.

Still open

- causal use of decoded directions;
- whether the model performs Bayesian updating;
- which coordinate system is most intrinsic.

NEXT QUESTION

Can predictive state be defined directly from observable futures, without choosing HMM coordinates first?

Full primary source: Shai et al., “Transformers Represent Belief State Geometry in their Residual Stream”.